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SolaraPay Case Study

Should the fund invest USD 4 million in a PAYGo solar company? (fictional worked case)

AUTHOR & IDEATION

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Case summary

SolaraPay Ltd and the Republic of Kivara are fictional. All figures, including the 24-month portfolio history, are synthetic teaching data generated by a documented, seeded script (`tools/cases/solarapay.py`). Every number in this case comes from the verified case workbook `SolaraPay_Case_Model_v0.7.xlsx` , or from the model's independent Python twin, which reproduces that workbook exactly.

The decision. SolaraPay Ltd, a PAYGo solar company in the Republic of Kivara (currency: Kivara shilling, KVS; opening rate KVS 135 per USD), asks an impact fund for **USD 4.0 million of equity at a USD 8.0 million pre-money valuation** (a 33.3% stake). In parallel it asks a local bank for a **KVS 4.5 billion receivables facility**. The fund's investment committee must decide: **Go, Conditional Go, or Stop?**

Recommendation: CONDITIONAL GO. The business can create value, but management's plan overstates repayment, recoveries and covenant headroom. The investment works only with conditions on pricing, Tier 4–5 exposure, collections, recovery evidence and valuation (Section 9).

Key figure (calibrated Base unless stated)	Value
Revenue, Year 1 → Year 5	USD 8.1m → USD 51.2m
EBITDA margin, Year 5	16.4% (management plan: 18.6%)
Equity required (peak)	USD 9.0m, with no top-up needed in Base
Investor IRR / MOIC (USD)	29.0% / 3.6x (management plan: 33.1% / 4.2x)
Investor IRR, Downside	Total loss (exit equity zero)
Lowest trailing 3-month collection rate vs 70% covenant	70.0%: no headroom
Observed collection rate, last 12 months of history	69.3% , already below the proposed covenant
Investment readiness	5 of 23 gates met: not investment grade

1. Company and market

SolaraPay Ltd has sold solar products on pay-as-you-go terms in Kivara for two years. Its customers pay a down payment and then a daily rate through mobile money. The device locks remotely if payments stop and unlocks permanently at the end of the plan.

The company sells **Tiers 1–3** today, and plans to launch two larger **solar-inverter + lithium systems (Tiers 4–5)** for urban households and small businesses with an unreliable grid.

Tier	Product	Cash price (KVS)	Implied APR	Plan tenor	Plan sales mix
1	Pico solar kit (10 Wp)	6,500	106%	12 m	25%
2	SHS with TV (80 Wp)	39,000	50%	24 m	40%
3	Large SHS + DC fridge (300 Wp)	117,000	39%	30 m	22%
4	Inverter + lithium (~1.2 kWp / 5 kWh) — new	390,000	32%	36 m	9%
5	Inverter + lithium (~3 kWp / 10 kWh) — new	845,000	25%	48 m	4%

Volume plan:

Year	1	2	3	4	5
Units sold	9,000	18,000	28,000	36,000	42,000

Capital structure proposed

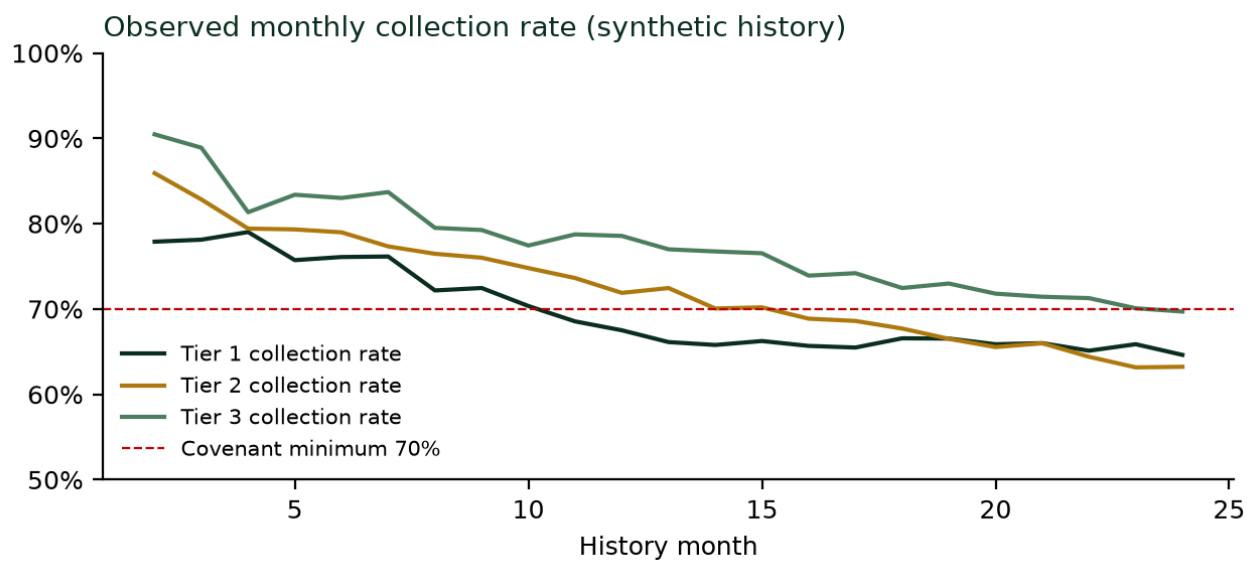
- Total initial equity of USD 9.0m (KVS 1,215m), including the fund's USD 4.0m.
- A USD 3.0m term loan (10%, 12-month grace, 36-month amortisation).
- A KVS 4.5bn local-currency receivables facility at 16%: up to 70–75% of eligible receivables, drawn as needed.

2. Step one: test the plan against SolaraPay's own data

Management's plan uses the default credit assumptions of the model:

- monthly default hazard: Tier 1 3.5%, Tier 2 2.6%, Tier 3 1.9%;
- collection on paying accounts: 88–90%.

Before projecting anything, the analyst loaded SolaraPay's **24-month portfolio history** into *Credit_Input*, and **18 monthly cohorts per tier** into *Vintage_Input*. Credit data mode was set to **Actual**.



What the history shows

1. Collections are deteriorating as the book seasons.

- The portfolio collection rate fell from **77.4%** in history year 1 to **69.3%** in history year 2.
- At the latest month, *Credit_Portfolio* reports a **66.5%** collection ratio, **17.0%** of receivables 30+ days past due, and an indicative ECL coverage of **36.5%**.

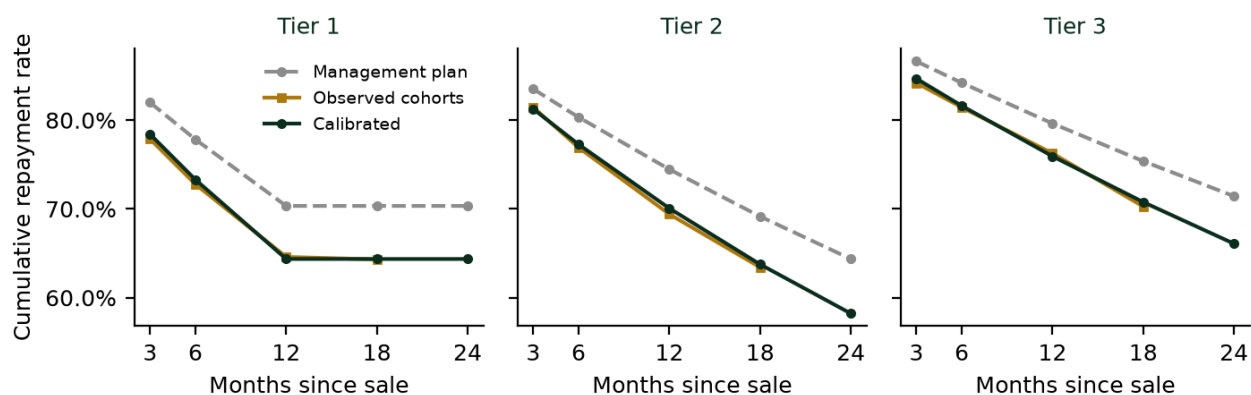
2. Cohorts repay below plan at every age. At month 12, observed cumulative repayment is:

Tier	Observed	Plan
Tier 1	64.6%	70.3%
Tier 2	69.4%	74.5%
Tier 3	76.3%	79.6%

3. Recoveries are almost absent. The observed LGD proxy is about **99%**: almost nothing is recovered from repossessed units. The plan assumes repossession of 20–70% of defaulted units, depending on tier.

4. There is no ownership evidence. No cohort is old enough to measure ownership at twice the tenor. Ownership-linked RBF is therefore impossible for now, and the model correctly pays zero under that design.

Vintage repayment curves: plan vs data vs calibration



Recalibration. The analyst re-estimated Tier 1–3 credit behaviour from the cohorts. The calibrated curves (green) sit on the observed points (gold):

Tier	Monthly default hazard: plan → calibrated	Collection on paying accounts: plan → calibrated
1	3.5% → 4.55%	88% → 86%
2	2.6% → 3.38%	88% → 87%
3	1.9% → 2.47%	90% → 89%
4–5	unchanged (no history)	unchanged (no history)

Lesson. The single most valuable hour of this analysis was spent on SolaraPay's own cohort data, not on the projections. A plan built on assumptions that the company's history already contradicts is not a base case.

3. Unit economics by tier

Calibrated Base (*Unit_Economics*):

Tier	Expected loss rate	Lifetime contribution (KVS)	LTV / CAC	Cash payback	Unit IRR
1	35.6%	1,092	2.4x	9 m	66%
2	41.8%	7,524	3.5x	17 m	37%
3	38.2%	33,897	5.5x	18 m	47%
4	27.1%	180,443	11.0x	17 m	67%
5	26.6%	416,975	14.9x	22 m	54%

Reading the table

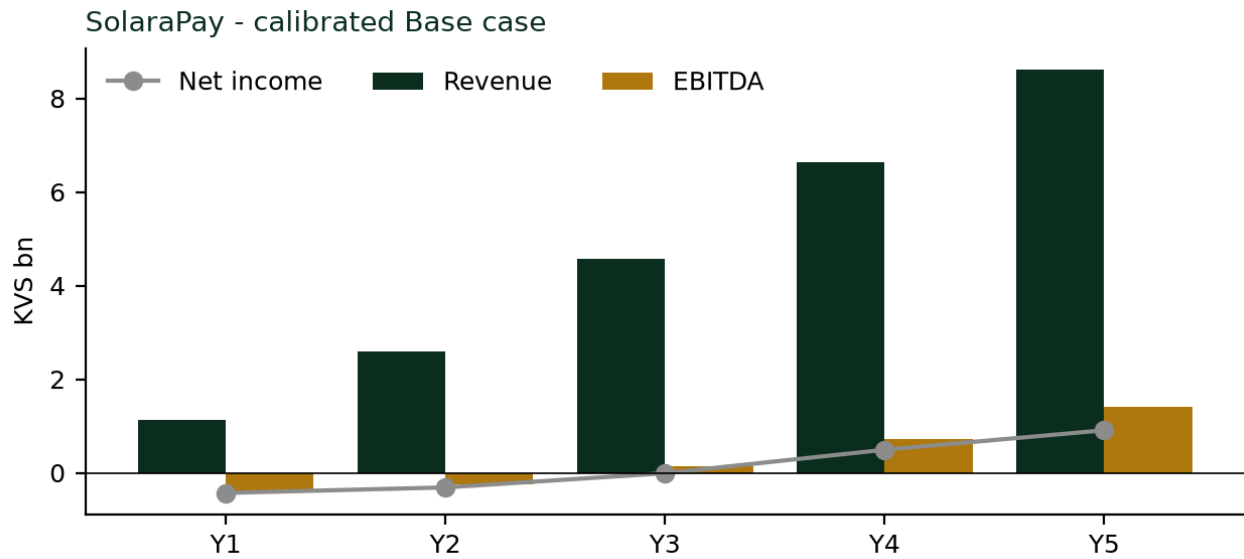
- Every tier still creates value at the unit level.
- **Tier 2, the volume driver, is the weakest product per unit.** It loses 41.8% of scheduled instalments and needs 17 months to pay back.
- **Tiers 4–5 look most attractive** (LTV/CAC 11–15x), but their credit assumptions have **no history behind them**.

Affordability (*Consumer_Risk*). The instalment exceeds the 10% payment-burden threshold for:

- **Tier 2** (13.0% of illustrative household income);
- **Tier 3** (13.2%);
- **Tier 4** (11.3%).

High burden is a leading indicator of the defaults seen in Tier 2. Tier 1's implied APR of 106% will attract regulatory and reputational attention.

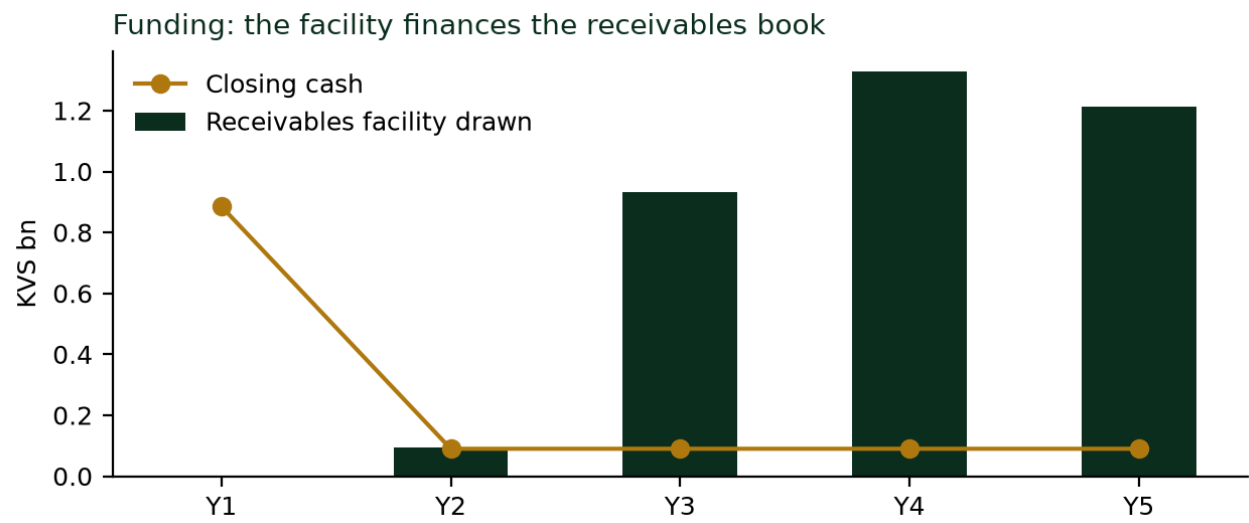
4. Calibrated projections



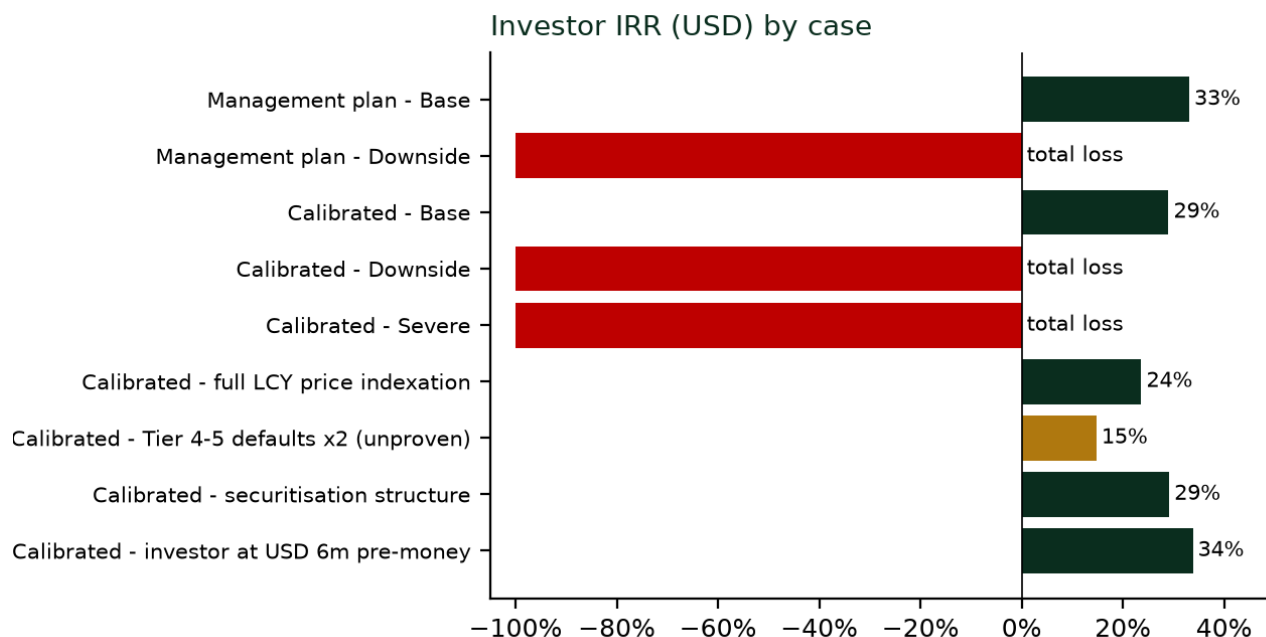
Calibrated Base	Y1	Y2	Y3	Y4	Y5
Revenue (KVS bn)	1.13	2.60	4.58	6.63	8.62
Gross margin	41.5%	46.8%	50.7%	54.0%	56.6%
EBITDA margin	-35.2%	-9.4%	2.9%	10.9%	16.4%
Net income (KVS bn)	-0.43	-0.31	-0.01	0.49	0.91
Net credit losses / revenue	37.5%	33.9%	31.3%	29.2%	27.5%
Collection rate	82.3%	76.8%	73.3%	71.5%	70.3%
30+ DPD / gross receivables	8.4%	11.4%	12.8%	13.7%	14.4%
Debt / book equity	0.54x	0.98x	2.51x	1.47x	0.65x

Highlights

- **EBITDA turns positive in month 29.**
- **Funding:** the USD 9.0m of initial equity is sufficient in Base. The receivables facility finances the growing book, peaking at total debt of KVS 1.45bn.
- **Debt service:** the annual DSCR is below 1.20x in Years 1–4. A cash-flow DSCR covenant is unsuitable for a growing PAYGo book; portfolio covenants are the right instruments (Section 7).



5. Stress tests



Case	Peak equity (USD m)	Y5 EBITDA margin	Y5 collection rate	Investor IRR	MOIC	Covenant-breach months
Management plan, Base	9.0	18.6%	73.0%	33.1%	4.2x	0
Management plan, Downside	9.0	3.7%	66.8%	total loss	0.0x	39
Calibrated, Base	9.0	16.4%	70.3%	29.0%	3.6x	0
Calibrated, Downside	9.0	1.3%	63.9%	total loss	0.0x	40
Calibrated, Severe	41.9	-25.4%	55.4%	total loss	0.0x	45
Calibrated Downside with full FX price indexation	9.0	14.2%	64.4%	23.6%	2.9x	38
Calibrated, Tier 4–5 defaults ×2	9.0	10.3%	63.8%	14.8%	2.0x	38
Calibrated, securitisation structure	9.0	16.4%	70.3%	29.1%	3.6x	0
Calibrated, investor at USD 6m pre-money	9.0	16.4%	70.3%	33.8%	4.3x	0

What the stress tests say

- 1. The Downside wipes out the equity.** At the Year 5 exit, net debt exceeds six times the (much smaller) EBITDA, so exit equity is zero. No new equity is needed during the five years, but the stake is worth nothing at exit.
- 2. Pricing power is the strongest mitigant.** Indexing new-contract prices fully to the currency turns the Downside from a total loss into a 23.6% IRR.
- 3. Tier 4–5 credit is the largest unproven risk.** Doubling their default hazard halves the IRR, to 14.8%, and triggers 38 months of covenant breaches.
- 4. Structure and price matter less than credit,** but they matter. Securitisation barely changes returns. A USD 6m pre-money lifts the IRR to 33.8%.

6. Valuation and returns

Metric (calibrated Base)	Value
DCF enterprise value at start (22% discount rate)	USD 6.1m
Exit equity value, 100%, end Year 5 (6x EBITDA less net debt)	USD 42.9m
Investor stake	33.3%
Investor IRR / MOIC (USD)	29.0% / 3.6x

Why the DCF is low. The DCF value is below the USD 8m pre-money, and its terminal value exceeds 100% of the enterprise value. Both are typical of a PAYGo company in growth. Every sale on credit is an investment in working capital, so five-year free cash flow is negative and all the value sits in the terminal year.

The fund should therefore:

- treat the price as a bet on the exit multiple and on credit quality;
- negotiate protection: a lower pre-money, a ratchet, or staged tranches tied to cohort performance.

7. Lender view

Test (calibrated Base, projection)	Result
Trailing 3-month collection rate, minimum 70%	Lowest 70.0%: no headroom
Receivables at risk, maximum 15%	Highest 14.4%
30+ DPD, maximum 25%; 90+ DPD, maximum 18%	Within limits
Debt / book equity, maximum 3.0x	Highest 2.51x (Year 3)
Annual DSCR, minimum 1.20x	Below in Years 1–4
Borrowing base (30 DPD eligibility, 70–75% advance)	Facility always within the base

The projection only just passes. On **actual data**, the latest trailing collection ratio (66.5%) already **breaches** a 70% covenant.

A lender should

- either set the collection covenant at a level the book can meet (for example 65%, with a cure period and a collections improvement plan);
- or delay the facility until the collection trend reverses.

The lender should also require monthly PERFORM-style reporting: collection rate, receivables at risk, and cohort repayment curves.

8. Benchmarks and calibration

Benchmark_Compare and *Calibration* set the calibrated Base against graded records of real companies. Peer coverage is thin (one or two companies per ratio), so these are **anecdotal checks, not norms**.

Ratio, Year 5	SolaraPay	Reference	Read-across
Net margin	10.5%	M-KOPA FY2024 $\approx$ 2.2% (provisional : revenue figures conflict)	4.8x the scale reference. Justify the cost and credit assumptions.
Expected credit losses / financing revenue	1.28x	M-KOPA FY2024 $\approx$ 0.38x	SolaraPay's credit losses are heavy relative to its financing income
Return on equity	49.8%	M-KOPA FY2024 $\approx$ 11% (equity derived from total assets – liabilities)	Optimistic relative to a scaled peer
Collection rate	70.3%	ESMAP sector reference $\approx$ 62% (2023, search snippet)	Within 10 points of the sector reference after calibration

9. Recommendation and conditions

CONDITIONAL GO: invest USD 4.0m, subject to the following conditions.

1. **Pricing policy.** Contractual or board-approved indexation of new-contract prices to the KVS/USD rate (the model's *pass-through* set to 100%). This is the most effective Downside mitigant identified.
2. **Tier 4–5 pilot cap.**
 - Tiers 4–5 limited to their planned 13% of units, released in tranches only after 12 months of cohort data confirm repayment within 10% of plan.
 - Separate portfolio-at-risk reporting for these tiers.
3. **Collections plan before the facility.**
 - A written collections improvement plan with monthly targets.
 - The facility covenant set at a level the book can meet (65% trailing collection with a cure period), stepping up as performance improves.
4. **Recovery assumptions.** Repossession and resale rates cut to observed levels, until a six-month repossession programme proves otherwise.
5. **Affordability redesign.** Restructure the Tier 2–4 price plans (longer tenors or higher deposits) so the instalment stays within 10% of target-segment income. Validate incomes with customer surveys.
6. **Valuation protection.** Negotiate the pre-money towards USD 6m, or a ratchet tied to Year 2 cohort performance.
7. **Data and validation.**
 - Monthly updates of *Credit_Input* and *Vintage_Input*, and ownership-at-2x tracking from launch.
 - Independent review of ECL and accounting (gate 19).
 - Workbook tested in Excel (gate 17).
 - Management sign-off (gate 2).

Readiness today: 5 of 23 gates met. These are model integrity, covenant compliance in Base, positive unit contribution, credit engine on actual data, and data reconciliation. The case is **not investment grade**; the conditions above close the most material gaps.

10. Questions for learners

1. Reset the credit assumptions to management's plan (*Products*: default hazard and collection rate). Which outputs change most, and why does the investor IRR move by about four points?
2. Set the FX pass-through to 100% in Base. What happens to affordability on *Consumer_Risk*, and what does that imply for defaults?
3. Lower Tier 2's tenor from 24 to 18 months with the same cash price. How do the payment burden, the expected loss rate and the unit IRR move?
4. Switch the RBF design to repayment-linked (mode 2), then to ownership-linked (mode 3). Why does mode 3 pay nothing, and what data would SolaraPay need to collect?
5. Set the borrowing-base eligibility to 90 DPD on *Credit_Assumptions*. Why does Base hardly change, while a lender would still refuse it?
6. Draft the collections covenant you would accept as a lender, using the 24-month history. Justify the level and the cure period.

Appendix — Reproducing this case

```
python tools/build_shs_model.py --case solarapay --snapshot      # builds the case workbook
python tools/qa_evaluate.py <case workbook>                    # evaluates every formula
python tools/case_exhibits.py <evaluated workbook> <cases.json> # regenerates the numbers
and figures
```

Synthetic data. The history is generated with the calibrated behaviour plus random noise (seed 20261002): 24 months of portfolio data for Tiers 1–3 and 18 cohorts per tier with checkpoints at M3–M24. It is teaching data only.

Known model point. The KPI "first month with positive operating cash flow" reads month 1 in this case. That is an artefact: the opening inventory is bought on 60-day supplier credit. It is not used here, and is logged for correction in the next model version.